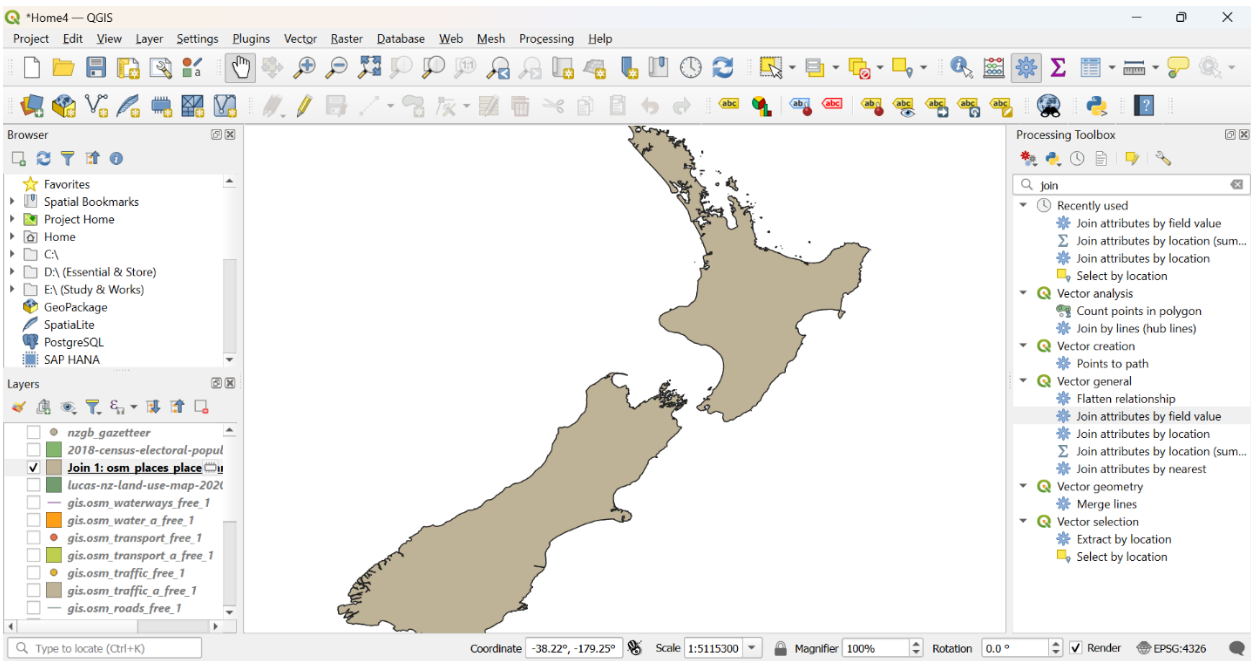
At-Home Activity 4

Started a new project at QGIS and loaded the following datasets:

1. Open Street Map for New Zealand
2. 2018 Census Electoral Population Meshblock 2020
3. Lucas NZ land use map
4. Place Name Data (gazetteer)

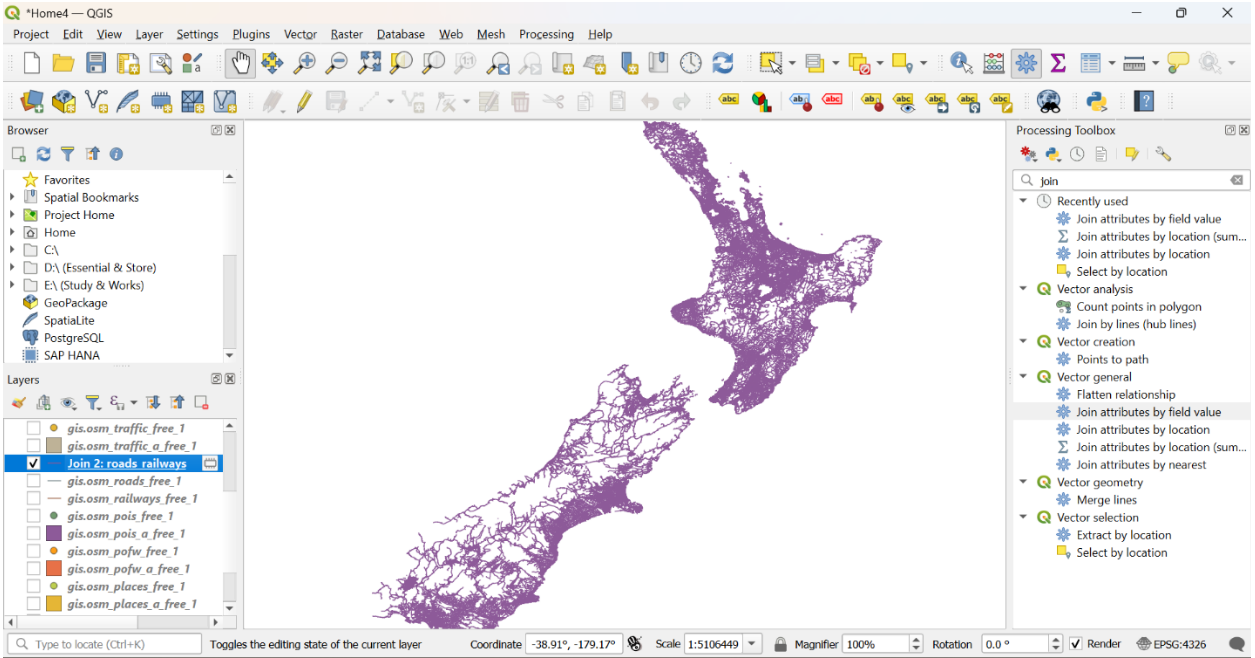
4: Join Attributes by Field Value

Join 01: Using Join Attributes by Field Value, joined osm_places_a_free and nzgb_gazetter layers by a common attribute ‘name’. So now, the place name attributes from the NZGB data is connected to the polygon data from OSM. Below is the view of the output layer for join 1.



Join 1

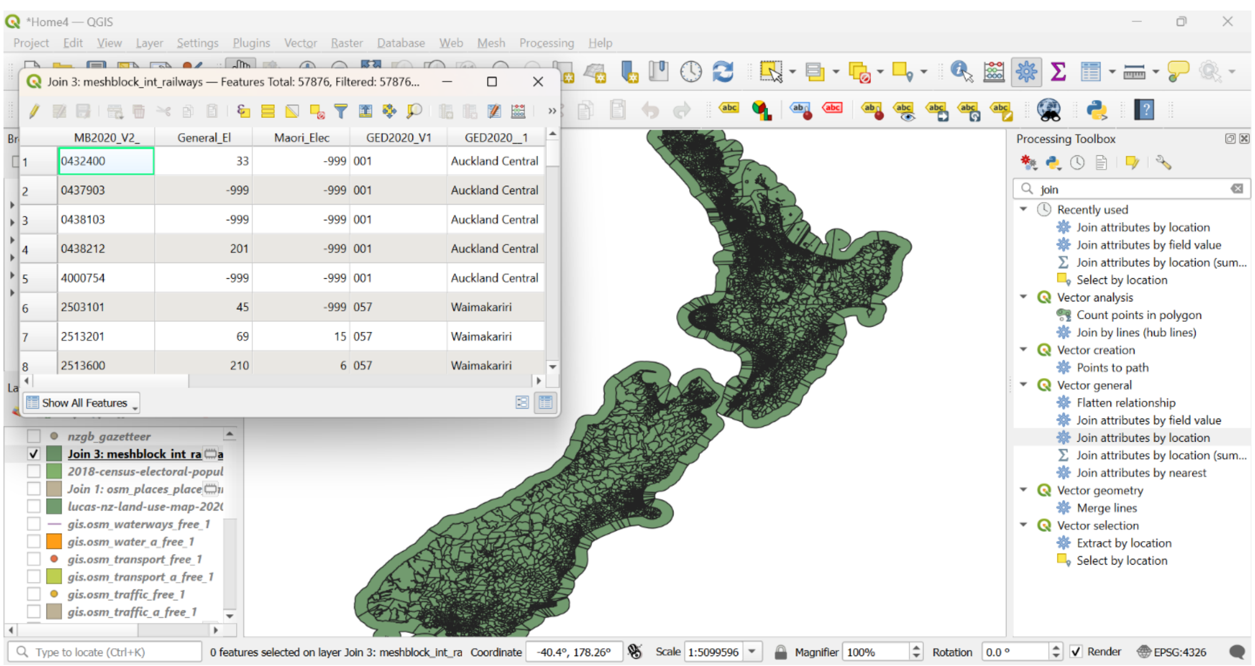
Join 2: To find out more meaningful connections, joined osm_roads_free_1 and osm_railways_free_1 using Join Attributes by Field Value by ‘name’ to join all the roads and railways. Here is the view of the output layer for join 2:



Join 2

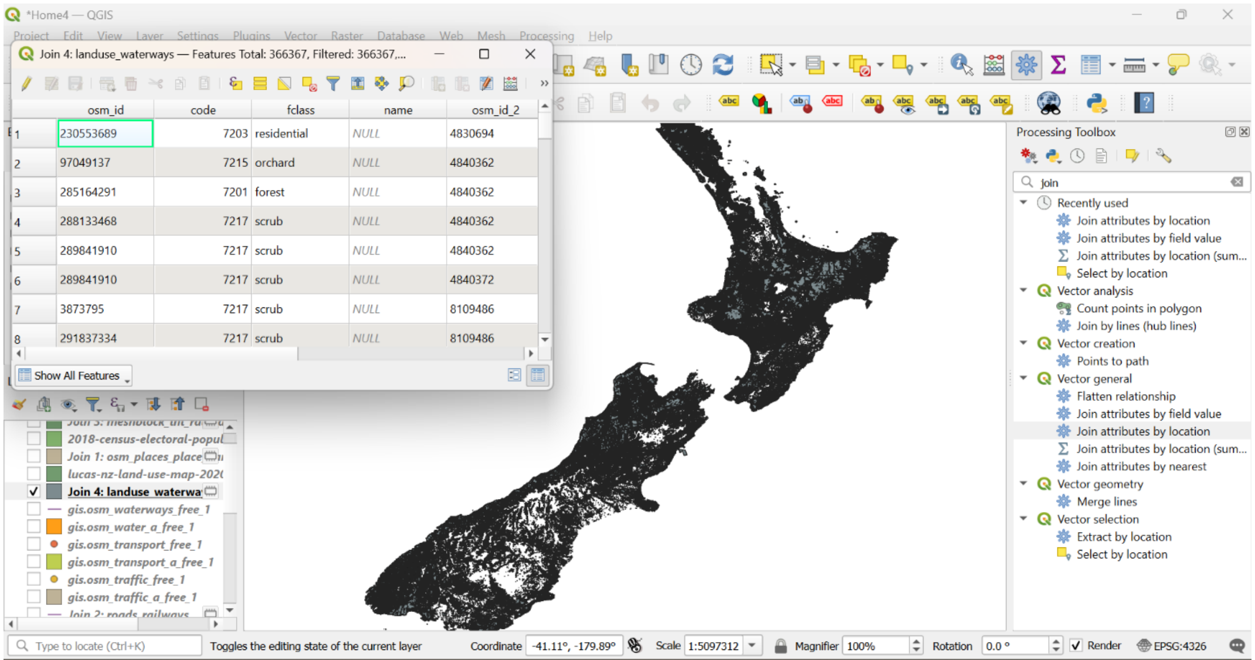
5: Join Attributes by Location

Join 3: To create a layer that will show the population for all meshblocks along railway lines used Join Attributes by Location tool. In as 2018 census population layer, to as osm_railways_free_1 and geometric predicate: intersect used to perform the join operation.



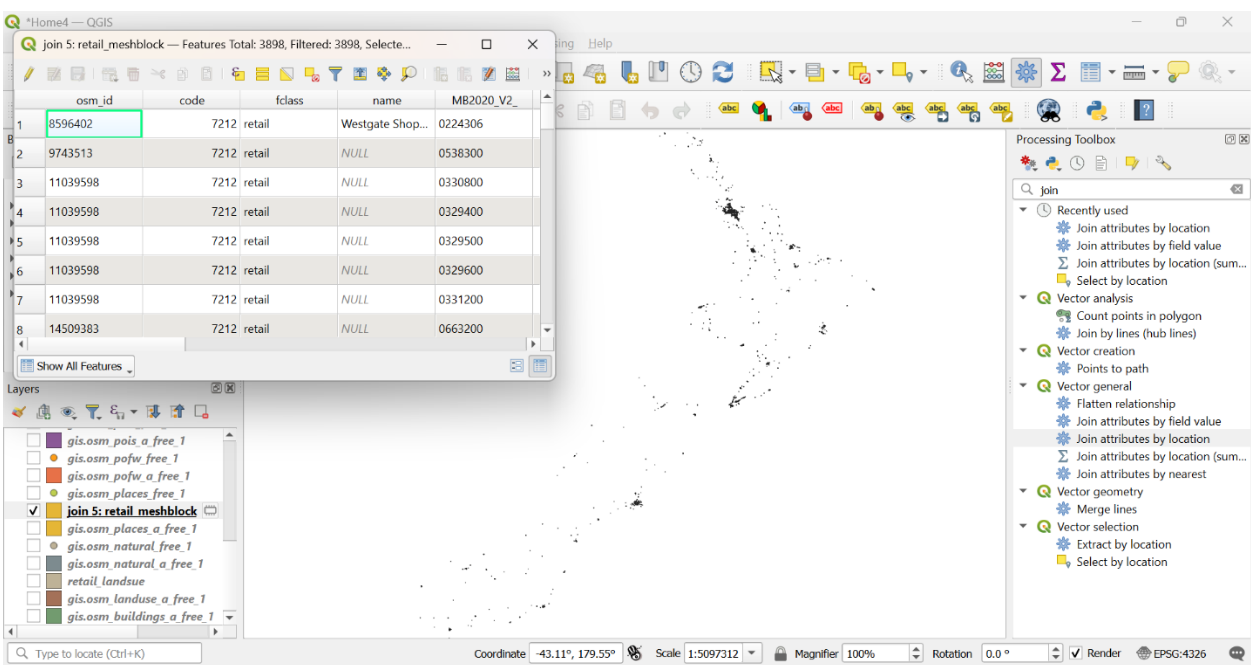
Join 3

Join 4: Another join operation performed by Join Attributes by Location using the layers landuse and waterways, geometric predicate: intersect to find out the land use types are intersecting waterways. Here is the output view:



Join 4

Join 5: Then, to find out populations within retail areas, performed another Join Attributes by Location tool using within predicate. After running the operation found the below map:



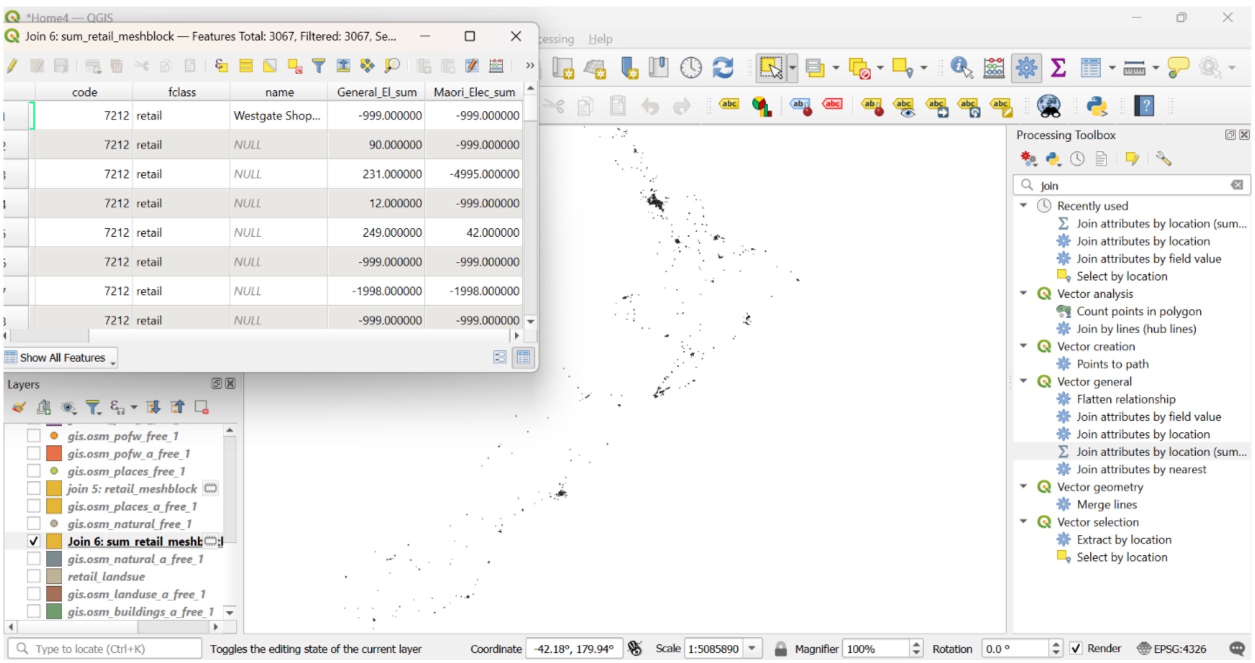
Join 5

6: Join Attributes by Location (Summary)

Join 6: Now, to find out the total population off all retail areas across New Zealand the tool Join Attributes by Location (Summary) is used. The parameters used:

- In: retail_areas
- To: 2018-census-electoral-population-meshblock-2020
- Geometric predicate: are within
- Field to Summarize: General_El, Maori_Elec
- Summaries to calculate: Sum
- Run

Then, found the following view and attributes:

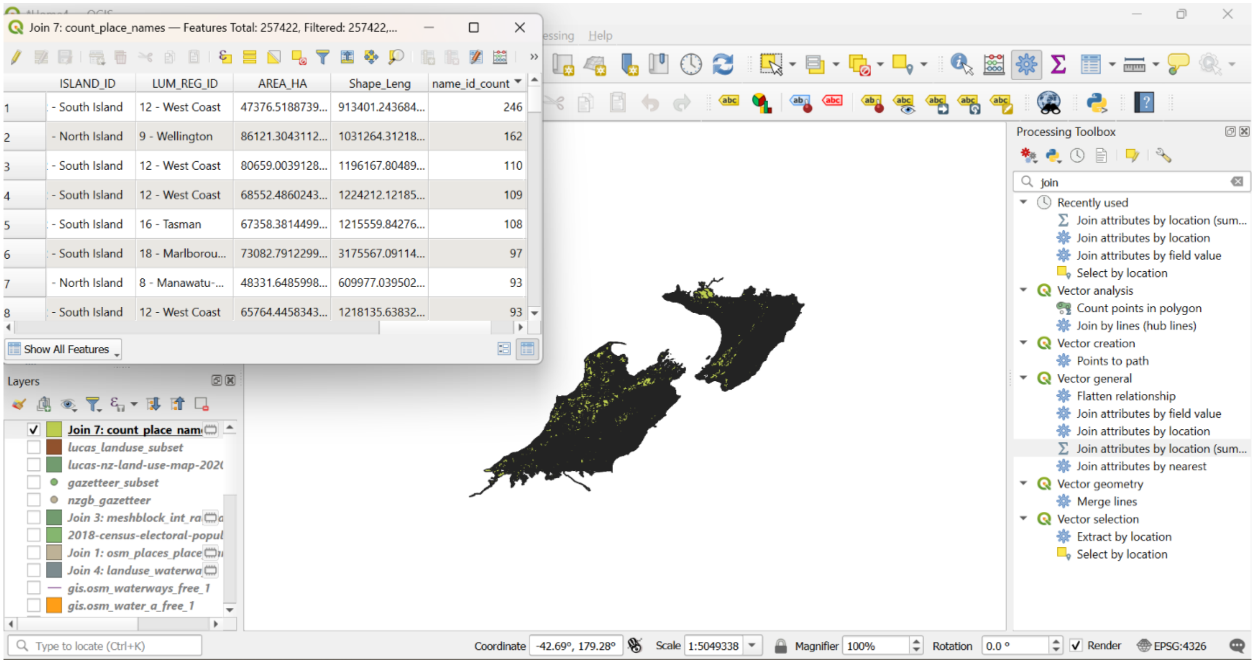


Join 6

Join 7: This time, have to find out which land use has the greatest number of place names in New Zealand. To do that selected a particular area to reduce complexity and time, opened the tool Join Attributes by Location (Summary) and performed following steps:

- In: nzgb_gazetter
- To: osm_landuse_free_a
- Geometric predicate: intersect
- Field to Summarize: osm_id
- Summaries to calculate: count
- Run

Then, found the desired below results:



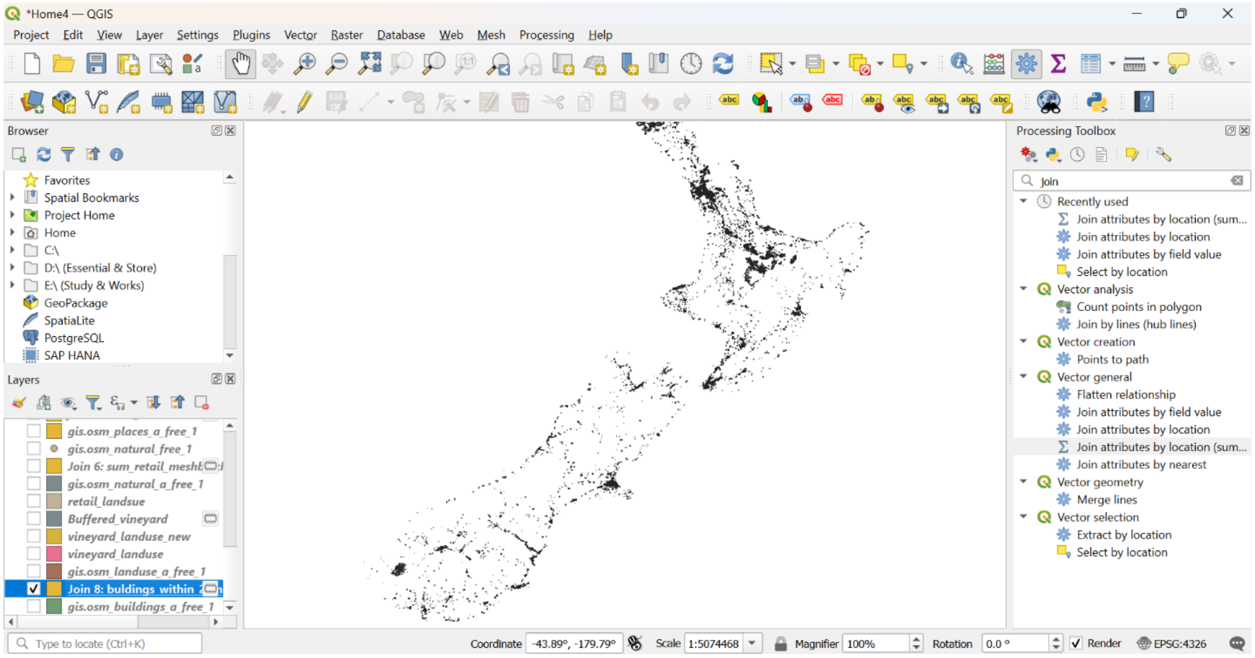
Join 7

Join 8: Finally, to find the total area of buildings that are within 2km of a vineyard in New Zealand, selected all the 'vineyard' values using Select Features by Expression ("fclass" = 'vineyard') and saved as vineyard_landuse name. As there is no predicate like within 2KM, had to create a buffer of 2000 meters of the new layer. For that selected the CRS 2193 to use meter as distance value. Then created the buffer using Buffer> Input Layer: vineyard_landuse> Distance: 2000 meters> Segments: 30> Run and got the buffered layer of 2KM of vineyard_landuse layer.

Next, used Join Attributes by Location (Summary) step by step as below.

- In: osm_buildings_a_free
- To: Buffered_vineyard
- Geometric predicate: intersect
- Field to Summarize: osm_id
- Summaries to calculate: sum
- Run

Found the total area of buildings that are within 2km of a vineyard in New Zealand as below:



Join 8

To answer all the questions, some of them required much time to perform operation, to reduce the complexity and time, used the following techniques:

- Create Spatial Index
- Created some subset layers

These strategies reduced the required time and complexity of performing operations.